

AJAY VENKATESH

Brooklyn, NY

+1 516 585 9013 av3855@nyu.edu linkedin.com/in/ajay-venkatesh-32013b17b github.com/ajayv-19

Education

New York University | Master of Science in Computer Engineering (GPA - 3.9/4)

Aug 2024 - Aug 2026

Experience

Amazon

May 2025 - Aug 2025

Software Development Engineer Intern

Seattle, US

- Designed a secure UI Console architecture using Midway SSO, OAuth, and HELIS to enforce least-privilege, role-based access; presented the solution to the team and received design approval.
- Built serverless infrastructure using AWS CDK to deploy API Gateway, Lambda (Java 21), DynamoDB, and S3 with cross-account access; implemented CI/CD pipeline for multi-stage deployments.
- Developed the frontend using React 18, integrating AWS UI components, Midway AuthN, and HELIS AuthZ with role-aware views and filtering; delivered a UI that was 40% faster and significantly more secure than older version.
- Implemented backend APIs using Java Coral, Smithy models, Dagger DI, and CBOR serialization; improved response time by 35% and ensured reliability through comprehensive unit and integration testing.

New York University

Aug 2024 - May 2025

Software Developer (On-Campus)

New York, US

- Contributed to an AI-driven Remote Patient Monitoring System, awarded \$100K by the National Science Foundation, USA, employing React Native, TypeScript, and Cognito authentication for real-time health monitoring.
- Authored parallelized ETL pipelines in Python to convert unstructured NoSQL data into structured SQL using GCP Cloud Functions. Leveraged concurrent processing of data chunks, reducing transformation time by 60%.
- Developed a React website to host QuickSight dashboard handling 50K+ records, utilized Cognito, Amplify, and Lambda functions to establish login, role-based access control, and Stripe integration for payment gateway.

PricewaterhouseCoopers (PwC)

Aug 2021 - Aug 2024

Senior Software Engineer

- Built a B2B responsive Dynamics 365 Marketing application for handling acquisition, creating deals, and onboarding, with capabilities like external data importing, data visualization using Power BI, and email tracking.
- Constructed a Cognitive Neural model using Azure ML to extract data from cards, integrated with Power Automate via API, reducing manual data processing time by 75%.
- Mentored 2 interns, conducted thorough code reviews for a team reducing bug occurrences by 25% and enabling a faster deployment cycle, aligning with agile methodologies.
- Engineered and optimized serverless database structure that improved data storage, retrieval, and processing workflows; achieved a 20% reduction in latency across diverse digital environments for seamless operations.
- Analyzed business goals and customer requirements to synthesize a comprehensive product roadmap and solution architecture for Microsoft Dynamics 365 B2B Health Application.
- Designed 6 targeted security protocols using Azure security groups that ensured appropriate control over logged-in users' privileges by assessing role hierarchies; fortified data protection measures and eliminated unauthorized access.
- Built a B2B responsive Dynamics 365 Marketing application for handling acquisition, creating deals, and onboarding, with capabilities like external data importing, data visualization using Power BI, and email tracking.
- Produced and integrated 5 Power BI dashboards to monitor and analyze data, establishing multi-level data filtering and security rules to access data, improving visualization and analysis efficiency.
- Composed client- and server-side custom configurations for CRM using JavaScript, plugins, custom workflows, webhooks, and business rules, resulting in a 35% faster server response.

Projects

Advanced Expense Tracker (Java Desktop Application) | Java, JavaFX, JDBC, SQLite, JFreeChart

- Architected a modular Java application supporting both individual and group expenses with secure login, leveraging OOP principles and design patterns (MVC and DAO) for separation of concerns.
- Implemented concurrent transaction handling and real-time cost splitting using multithreading and synchronized blocks to ensure thread-safe database updates in group expense scenarios.

Skills

Programming Languages: C#, Python, JavaScript, TypeScript, SQL

Web Development: HTML, CSS, React, Node.js, Express.js, Spring Boot

Database: Dataverse, RDBMS (MySQL, PostgreSQL), NoSQL (MongoDB, JSON)

Cloud: Azure (AI Services, Logic Apps, Databricks, Function Apps, Storage Blobs, Security Groups), AWS (EC2, S3, Amplify, Lambda Function, API Gateway, Cognito, IAM, AppSync, QuickSight, Q, SageMaker)

Tools: Git, Azure DevOps, Visual Studio, Dynamics CRM 365, ERP, XRM Toolbox, .NET, Power Automate, Power BI

Methodology: Agile, Waterfall

AI/ML: Machine Learning, Deep Learning, Big Data, MLOps, Principal Component Analysis

Certifications

AWS Cloud Practitioner

Microsoft Solution Architect Expert

Microsoft Developer Associate

Azure Data Scientist Associate

Power BI Data Analyst Associate

Azure Data Fundamentals